Machine Design (brouillon)

Yzh & Mattx

1 – Homework 2

1.1 – Equations

For the fluids :

- At the start, we have air and a fuel. And after the combustion, we know that we obtain CO_2, H_2O, N_2, O_2.
- We can study other case (the initial one is with CH_4)

État	p (Pa)	T (K)	h (J/kg)	s (J/kgK)	e (J/kg)
1	$\boldsymbol{p_1}$	$\boldsymbol{T_1}$	$h(p_1, T_1)$	$s(p_1, T_1)$	$\Delta h_{01} - T_0 \Delta s_{01}$
2	$p_1 r_c$	$T_1 \left(\frac{p_2}{p_1}\right)^{\frac{m_C - 1}{m_C}}$	$h(p_2, T_2)$	$s(p_2, T_2)$	$\Delta h_{02} - T_0 \Delta s_{02}$
3	$p_2 k_{cc}$	$\boldsymbol{T_3}$	h_3	$s(h_3, T_3, \text{gaz})$	$\Delta h_{03} - T_0 \Delta s_{03}$
4	p_1	$T_3 \left(\frac{p_4}{p_3}\right)^{\frac{m_T - 1}{m_T}}$	$h(p_4, T_4, \text{gaz})$	$s(p_4, T_4, \text{gaz})$	$\Delta h_{04} - T_0 \Delta s_{04}$

(State 1)

For the initial state, we already know p_1 and T_1, and the fluid is the air. Then we can find the values of $h_1 = h(p_1, T_1)$ and $s_1 = s(p_1, T_1)$, but also :

$$e_1 = (h_1 - h_0) - T_0(s_1 - s_0)$$

where $h_0 = h(p_0, T_0)$ and $s_0 = s(p_0, T_0)$, as we know p_0 and T_0.

(State 2)

It is also the air. As we know the pressure ratio r_c, and as the transformation $1 - 2$ is with a compressor, we can say that :

$$r_c = \frac{p_2}{p_1} \qquad \rightarrow \qquad p_2 = p_1 r_c$$

And we also have $\eta_{\text{Pi,C}} = \frac{m_C}{m_C - 1} \frac{\gamma - 1}{\gamma}$, so $m_C = \frac{1}{1 - \frac{\gamma - 1}{\gamma \eta_{\text{Pi,C}}}}$, and then $T_2 = T_1 \left(\frac{p_2}{p_1}\right)^{\frac{m_C - 1}{m_C}}$. For the rest, we just have to compute $h_2 = h(p_2, T_2)$, $s_2 = s(p_2, T_2)$ and :

$$e_2 = (h_2 - h_0) - T_0(s_2 - s_0)$$

(State 3)

For the combustion, we have a fuel, CH_4, such as $x_C = 1$ and $x_H = 4$. The chemical equation for the CH_4 is already given. We can directly write that $p_3 = k_{cc} p_2$, by the definition of k_{cc}. And T_3 is known. By the exercise, we have a formula for h_3, so we can calculate it by using that. But we have to find λ with an implicit equation, because $h_3 = h_3(\lambda)$:

$$\lambda = \frac{\text{LHV} + h_f - h_3}{m_{a1}(h_3 - h_2)}$$

Then, we find $\dot{m}_a, \dot{m}_g$ and $\dot{m}_f$ by using the following equations :

$$\dot{m}_g = \dot{m}_a + \dot{m}_f \qquad\qquad \dot{m}_a = \lambda m_{a1} \dot{m}_f$$

At the same time, we find $s_3 = s(p_3, T_3, \text{gaz})$ and :

$$e_3 = (h_3 - h_0) - T_0(s_3 - s_0)$$

(State 4)

As we know p_3 and T_3, and by knowing that $p_4 = p_1$ (isobaric transformation) :

$$\eta_{\text{Pi,T}} = \frac{m_T - 1}{m_T} \frac{\gamma}{\gamma - 1} \qquad \rightarrow \qquad m_T = \frac{1}{1 - \frac{(\gamma-1)\eta_{\text{Pi,T}}}{\gamma}}$$

And then $T_4 = T_3 \left(\frac{p_4}{p_3}\right)^{\alpha}$ where $\alpha = \frac{m_T - 1}{m_T}$. Then :

$$h_4 = h(p_4, T_4, \text{gaz}) \qquad s_4(p_4, T_4, \text{gaz}) \qquad e_4 = (h_4 - h_0) - T_0(s_4 - s_0)$$

(Gaz)

To determine the moles of G :

$$x_G = \frac{n_G}{n_{\text{tot}}}$$

(Rest)

For the work :

$$W_{\text{mcy}} = w_T - w_C = (h_3 - h_4) - (h_2 - h_1)$$

We can find $\dot{m}_g$ by using P_e, such as $\dot{m}_g = \frac{P_e}{W_{\text{mcy}}}$, and then $\dot{m}_a$ and $\dot{m}_f$. And also :

$$\eta_{\text{cycle}} = \frac{P_e}{\dot{m}_f \text{ LHV}} = \frac{\dot{m}_g W_{\text{mcy}}}{\dot{m}_f \text{ LHV}}$$